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Feasibility Study on Cloud Communication Operation for an Interventional Surgery Robot

Yangming Guo[1], Shuxiang Guo[1, 2*], Cheng Yang[1]
[1]*Key Laboratory of Convergence Medical Engineering System and Healthcare Technology,*
The Ministry of Industry and Information Technology, School of Automation, Beijing Institute of Technology, No.5,
Zhongguancun South Street, Haidian District, Beijing 100081, China
[2]*Faculty of Engineering, Kagawa University, 2217-20 Hayashi-cho, Takamatsu, Kagawa 760-8521, Japan*
E-mails: guoshuxiang@bit.edu.cn; guoyangming@bit.edu.cn
*Corresponding author

Abstract **- The use of robotic technology in the process of vascular interventional surgery has become a common method of treating cardiovascular diseases. A major advantage of the vascular interventional surgery robot is that it can isolate the doctor from the operating room and greatly reduce the radiation damage to the doctor's body. A remote vascular intervention robot system based on cloud communication was developed for medical applications. In the remote operation of interventional therapy, the doctor operate the master controller, then master controller sends control information to the cloud server, which in turn sends control information to the slave operator and controls it to perform the operation. Experimental results show that the cloud communication system can meet the feasibility requirements of remote vascular interventional surgery.**

Index Terms - Interventional Surgical Robot, Cloud Communication, Cloud server, Remote surgery.

I. INTRODUCTION

Cardiovascular and cerebrovascular diseases affect millions of people worldwide. In 2015, an estimated 18 million people died of cardiovascular and cerebrovascular diseases. Vascular interventional surgery is considered to be very effective for the treatment of such diseases method. However, traditional vascular interventional surgery has many limitations. First of all, during the operation, patients and medical staff are exposed to high doses of X-ray radiation. Second, performing surgery in a small operating space may lead to misoperation. Third, surgeons often become fatigued by having to wear heavy lead clothing to reduce their radiation. Therefore, the advent of vascular interventional robots can solve all these problems.

In recent years, numerous studies have been conducted in relation to vascular intervention robotic systems. The robotic-assisted minimally invasive surgical robotic system is now able to perform multiple types of surgery, such as skull base surgery, mandibular reconstruction surgery, vascular surgery and so on. And the current trend of medical robots is to develop telephoto medical robots. It describes a variety of remote operation medical robots, such as the most prestigious Da Vinci robots. Teleoperated Medical robot consists of multiple devices that require the transfer of information between each other a distributed modular architecture has been developed that can be applied to a variety of surgical robotic systems, providing a reference for telemedicine robots.

In the past ten years, several vascular interventional surgical robots have been developed. For example, Hansen Medical has developed the Sensei robot system, in which the doctor manipulates the manipulator with the help of the imaging system, so that the slave system can be controlled. The Sensei robot system provides four operating ratios, and doctors can choose different control modes according to their needs [1]. The Amigo remote surgical robot was designed and developed by Catheter Robotic Inc. and has been used in clinics. More than 200 patients have used the robot. And the company has been constantly upgrading the robot. Currently, Amigo RCS robot is currently the company's latest robot [2]. Stereotaxis Inc. developed the Stereotaxis Niobe robot system in 2002, which is a magnetic navigation intervention system. Stereotaxis Inc. has been committed to the improvement of the system for a long time and updated its product to Niobe ES MNS in 2012 [3]. Corindus Vascular Robotics has developed the Corpath 200 system, which is a vascular interventional robotic system. The system is an open system, and the catheter can be selected according to the doctor's operation requirements [4]. In the school's research institutions, many outstanding achievements have also been made. Professor Guo of Beijing Institute of Technology has developed a new type of vascular interventional surgery robot system that combines a master-end controller and a slave-end manipulator to realize remote operation [5] [6] [7] [8]. Professor Ganji built a catheter navigation platform to control cardiac radiofrequency catheters and completed related experiments. Professor Fu developed and designed a vascular interventional surgery robot system. The robot system is master-slave operation and consists of the following four parts: master handle, mechanical auxiliary equipment, 3D guidance image and electromagnetic sensor (mainly used for Positioning catheter) [9].

However, most interventional surgery robots can only be operated outside the operating room, which severely limits the operability space of doctors, so interventional surgery robots that can be controlled remotely become particularly important.

This paper proposed a novel cloud communication operation for an interventional surgery robot, which allows doctors to operate surgical robots beyond a long distance, greatly expanding the scope of application of interventional surgery robots [10] [11]. The result of related experiments proves that the operation scheme is theoretically feasible. And

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it can be further optimized to meet the needs of remote vascular intervention surgery robots

II. Interventional Surgical Robot System

One of the main problems solved by the remote vascular interventional surgical robot is that the doctor cannot remotely control the movement of the catheter guide wire [12]. Based on the research of vascular interventional surgery robots developed by some domestic and foreign companies and research institutions, this paper designed a remote vascular interventional robot system based on cloud communication. The vascular surgery robot can enable a doctor to operate a patient even if he is not in the operating room of the patient [13] [14].

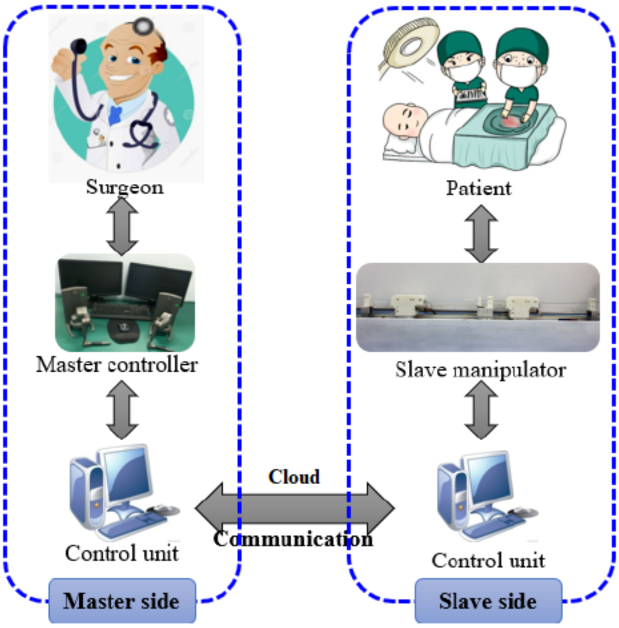

Fig.1 Schematic diagram of the system [15]

The vascular intervention surgery robot system includes a master controller [16] [17], a slave operator [18] [19], and a cloud server (that is, a cloud control system). The doctor manipulates the master controller to push, pull and rotate, and then the master controller sends the control information to the cloud server. The cloud server then sends the control information to the slave operator. After receiving the control information, the slave operator controls the movement of the guide wire and the catheter. Therefore, the decoupling of the master controller and the slave operator is realized, which provides the doctor with the possibility of remote operation [20].

A. The master controller

The purpose of using a vascular interventional surgical robot is to isolate the doctor from the operating room. Generally speaking, doctors operate the master controller outside the operating room, because there is no radiation outside the operating room, which is safer for the doctor, while the surgical robot replaces the doctor in the operating room to perform operations on the patient's body. Since the doctor does not directly touch the patient's body, it is

Fig.2 Our Master Controller [21]

particularly important that the master controller installed outside the operating room can accurately provide force feedback to the doctor. Most of the current researches use motors to give force feedback to the doctor’s hands, but this seriously affects the degree of freedom of the master controller, and there are problems with high delay time and low force feedback accuracy. Therefore, the master controller needs to have the following functions:

1).The design of the master controller should conform to the doctor's operating habits. When the doctor operates the master controller, it should be like to operate the catheter directly.

2).The master controller should have the function of real-time sampling of data. When the doctor operates the master roller, it should immediately calculate the pushing distance, pushing speed, rotation angle, rotation speed and other information of the doctor's hand.

Two Geomagic Touch X devices were chosen as master controllers(one is used to control the catheter, the other is used to control the guide wire), which have general communication and charging ports, support all commonly used software, and are easy to install and deploy. Moreover, it has 6 degrees of freedom of position and 3 degrees of freedom of force feedback, which is sufficient to meet the operation requirements of doctors during surgery [22].

B. The slave operator

In the remote vascular interventional surgery operation, the catheter and the guide wire are in directly contact with the blood vessels of the patient's body, and the catheter and the guide wire are always in the patient's body during the operation. Therefore, stable, accurate and safe control of catheter and guide wire is required. Slave operator was designed to directly control the catheter and the guide wire. The slave operator includes two parts, one is used to hold the catheter, and the other is used to hold the guide wire, and the two parts are decoupled [23] [24]. Therefore, our slave operator can manipulate the catheter or the guide wire to complete linear and rotation operations, which was shown in Fig.3.

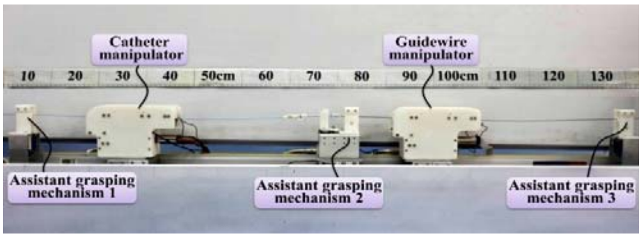

Fig.3 Slave operator [23]

III. The Cloud Communication Of The System

A. Network System Architecture

In this paper, a set of network system framework that is suitable for master-slave communication is designed to meet the requirements of communication in vascular intervention robot system. This network system architecture, which can be used for variety of vascular intervention surgery robot system, can easily connect the various components of the system. The TCP/IP protocol refers to the OSI architecture to a certain extent. The OSI model has a total of seven layers, from bottom to top are the physical layer, data link layer, network layer, transport layer, session layer, presentation layer and application layer. But this is obviously a bit complicated, so in the TCP/IP protocol, they are simplified to four levels, which was shown in Fig.5.

1).The services provided by the three layers of the application layer, presentation layer, and session layer are not very different, so in the TCP/IP protocol, they are merged into one layer of the application layer.

2).Since the status of the transport layer and the network layer in the network protocol is very important, they are regarded as two independent layers in the TCP/IP protocol.

3). Because the contents of the data link layer and the physical layer are similar, they are merged into one layer of the network interface layer in the TCP/IP protocol. The TCP/IP protocol with only a four-layer architecture is much simpler than the OSI with a seven-layer architecture. This is exactly the case. The TCP/IP protocol has higher efficiency and lower cost in practical applications..

B. Details of Network System Architecture.

A set of TCP / IP communication is designed. This communication establishes two paths so that transmission and reception do not interfere with each other. In the process of communication with each other, there is no waiting problem between the incoming and outgoing message, which satisfies the requirement of real-time transmission of the message in the vascular intervention robot system. In the process of communication, through the transmission of messages, the session will be diagnosed to judge if the session is disconnected. Due to the establishment of a communication diagnostic mechanism, when the communication is interrupted, the session can be reconnected in time.

In this paper designed a set of vascular intervention surgery robot system communication architecture, based on the TCP / IP communication method, so that this communication architecture can ensure the communication fluency and real-time capability. In master controller and slave operator, a server has been in the state of monitoring whether there is the active connection. Once receiving a signal that represents that other device actively try to connect to itself, it accepts the active connection and saves the socket and can manage the connection to the device. When the session between the devices is disconnected, the session can be reconnected because the connection to the device can be managed. The communication mode used in the communication architecture is asynchronous communication, which can ensure that so that there will be no communication jam. This communication architecture can be applied to a variety of vascular intervention surgery robot system to meet

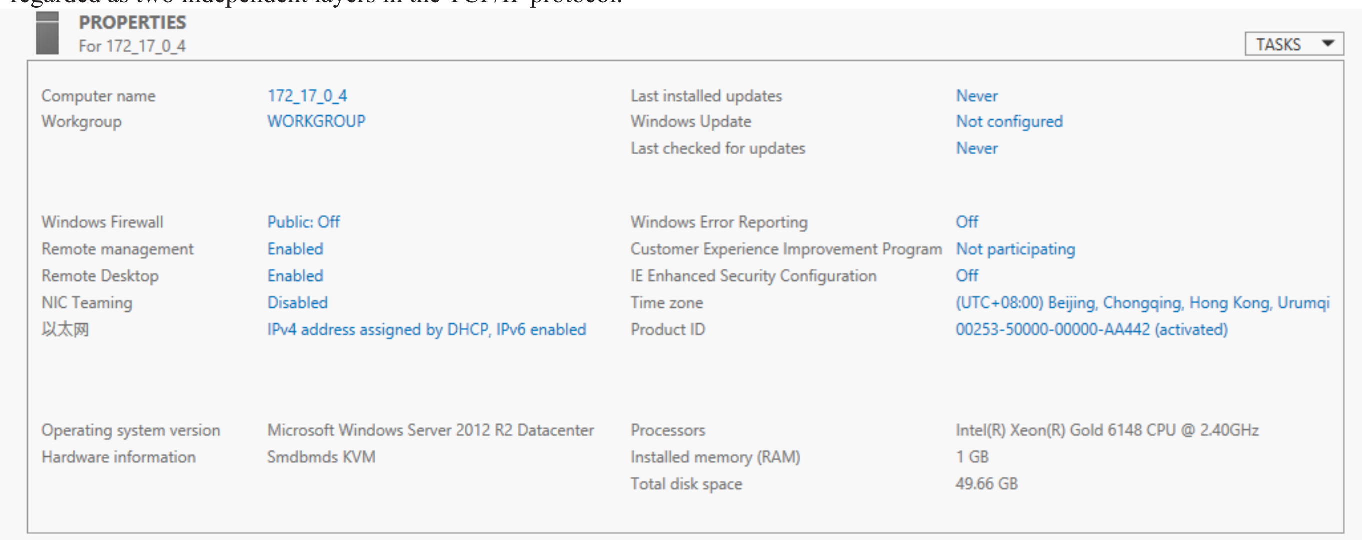

Fig.4 Developed Cloud Server

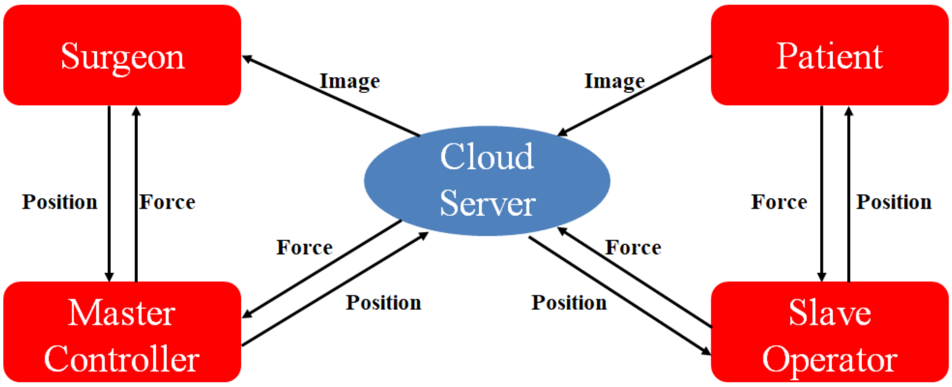

Fig.5 The Network System Architecture

its real-time operation and feedback that communication needs.

In the preliminary communication protocol, the control signal between the master controller and the cloud server occupies less memory, because only the doctor's action information is valid in the data sent by the master controller to the cloud server. Similarly, in the communication process between the cloud server and the slave operator, the useful information only contains the doctor's operation information. However, in the communication between the slave operator and the cloud server, the information is relatively rich, which can include angiography images in the operating room, real-time position images of catheters and guide wires, and these images will also be transmitted to the end of the doctor in real time. So in this part of the network communication, the bandwidth will be mainly used to transmit image data. At the same time, for security reasons, in case of a network link break, a spare network link needs to be prepared to prevent the network from being broken.

C. Cloud Server.

Windows server 2012 R2 Datacenter version was used as our cloud server, and the processor is Inter(R) Xeon(R) Gold 6148 CPU @ 2.40GHz, the installed memory(RAM) is 1.00 GB, the system type is 64-bit operating system.

Windows Server 2012 R2 is a new generation of Windows Server operating system based on Windows 8.1, providing enterprise-level data center and hybrid cloud solutions that are easy to deploy, cost-effective, application-focused, and user-centric. Windows Server 2012 R2 features include server virtualization, storage, software-defined networking, server management and automation, Web and application platforms, access and information protection, virtual desktop infrastructure, etc.

IV. Experimental Result And Analysis

In order to verify the real-time performance of the remote intervention surgical robot system, the cloud server is set up in Shanghai, and the master controller and slave operator are set up in Beijing, so as to calculate the communication delay from the master controller to the cloud and from the cloud to the slave operator.

The specific delay test method is that the master controller sends a data packet to the cloud server. The data packet contains the system time and position data of the master controller. After receiving the data packet, the cloud server immediately leaves the data packet intact It is sent back to the main controller without movement. After receiving the data packet from the cloud server, the master controller verifies whether the position data in the data packet is the position data sent by itself, and records the system time at the moment. The system time at this moment minus the system time in the received data packet, and divides the result by two to obtain the communication delay between the master controller and the cloud server. Due to the different degree of network congestion during a day, the delay data is obtained after multiple measurements and averaging.

After testing, when the master controller performs linear and rotating actions, the communication delay from the master controller to the cloud server is 25ms. Since the communication link from master controller to cloud and cloud to slave operator is the same, the communication delay from cloud to slave operator is also 25ms.

The speed of network communication involves the influence of network bandwidth and the number of routers, the above data is not optimal data. Because the existing cloud control system is a preliminary test system, only 1M of network bandwidth is selected, and the number of routes is not clear (the number of routers needs to be communicated with the cloud server provider to confirm). In the system after optimization, if it involves clinical trials and commercialization, we will change the performance of the network bandwidth and optimize the number of routers.

In addition, considering that the cloud server we are using now only has 1GB of installed memory, there is still a higher room for improvement in data processing and information transmission, so if we use multi-core processors, or use GPUs for our data Processing, the delay of the system can be further reduced.

Due to some irresistible factors, part of the experimental data of the system is not perfect. The simple control instructions may not meet the clinical needs of remote interventional surgery robots, but video and voice signals can be added later in the experiment. And then measure the transmission delay of these information. This experiment is of great research significance. By measuring the delay data caused by different server locations, different bandwidths, and different routing numbers, it can provide a lot of valuable information for building more reliable and faster remote surgery systems in the future.

V. Conclusions

This paper proposed a novel cloud communication operation for an interventional surgery robot. Through the use of cloud servers, doctors and patients can be separated in space to improve the utilization efficiency of high-quality doctors, and can fundamentally reduce the radiation damage received by doctors themselves. After the experiment, the communication delay between the master controller and the slave controller is 50ms, which is not suitable for vascular intervention surgery, so the next step is to reduce the communication delay between the master controller and the slave controller. In addition, the real-time information feedback of the slave operator is also very important in vascular interventional surgery, so the next step will focus on the real-time video information feedback from the slave operator to the master controller.

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